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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Fourth Semester B.E. (CSE) (Credit System) Degree Examinations

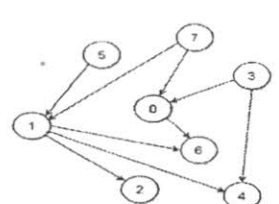
Supplementary Examinations - December 2025

21CS402/20CS402/21IS402/20IS402/21AM402/21CC402/20CC402 – DESIGN AND ANALYSIS OF ALGORITHMS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

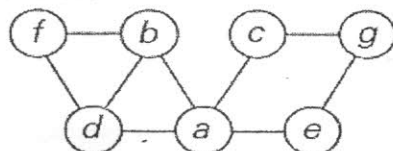
Unit – I		Marks	BT*	CO*	PO*
1.	a) List out the general plan for the analysis of non recursive algorithms and apply it for the element uniqueness problem of finding whether the elements of the given list are distinct or not.	8	L*3	1	1
	b) Illustrate the formal definition of Big-Oh(O) notation with example. Also prove that if $f_1(n) \in O(g_1(n))$ and if $f_2(n) \in O(g_2(n))$, then $f_1(n) + f_2(n) \in O(\max\{g_1(n), g_2(n)\})$	8	L2	1	1
	c) Algorithm: Riddle(A[0 ..n-1]) //Input: An arrayA[0 ..n-1] of real numbers if n = 1 return A[0] else temp ←Riddle(A[0 ..n-2]) if temp ≤ A[n-1] return temp else returnA[n-1] a.What does this algorithm compute? b.Set up a recurrence relation for the algorithm's basic operation count and solve it	4	L2	1	1
2.	a) Construct an algorithm for mergesort. Also state the Master's theorem and calculate the worst time efficiency of mergesort algorithm using the Master's theorem.	8	L3	2	1
	b) Explain the divide an conquer method. Outline an algorithm for binary search and derive its worst case time efficiency.	8	L2	2	1
	c) Define the Big Omega(Ω) and Big-Theta(θ) notations with example	4	L1	1	1
3.	a) Outline the algorithm for Quicksort. Apply Quick Sort algorithm on the following characters: Q, U, I, C, K, S, O, R, T Show the split positions clearly using the tree structure.	8	L3	2	1
	b) Explain the best case, worst case and average case analysis of a typical algorithm? Explain the same for sequential search algorithm.	8	L2	1	1
	c) Outline the brute force string matching algorithm.	4	L2	2	1
Unit – II					
4.	a) Explain the two methods for finding the topological order for any graph. Apply any of the methods to write the topological sort order for following graph.				
		8	L3	3	1
	b) Explain the algorithm for bottom up heap construction and apply heapsort algorithm for the following list: 28 15 45 6 17 32 3 12	8	L3	3	1
	c) Explain the three major variations of decrease and conquer technique with an example for each.	4	L2	3	1

P.T.O.

5. a) Apply the bottom up dynamic programming algorithm to the following instance of the knapsack problem. Capacity $W=6$

Item	Weight	Value
1	5	10
2	4	40
3	6	30
4	3	50

- b) Explain the Depth First search algorithm. Apply DFS algorithm on following graph to list the order in which vertices are visited and construct the DFS forest.



- c) Outline the Warshall's algorithm to find the transitive closure of a given directed graph.
6. a) Outline an algorithm for insertion sort and apply it to sort the following elements: A L G O R I T H M S. Also state its the worst case and best case time efficiency.
- b) Outline Horspool's String Matching Algorithm and apply the same for the following input: *Text* : NITTE UNIVERSITY *Pattern* : SIT
- c) Outline the algorithm for Comparison counting sorting technique.

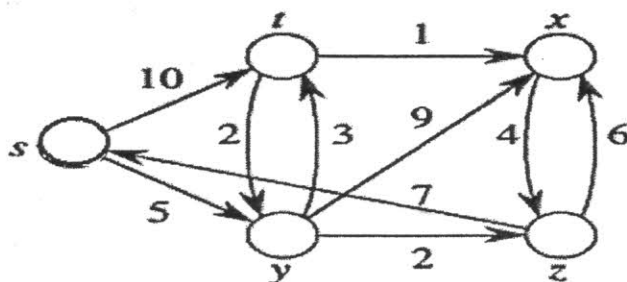
Unit – III

7. a) Explain huffman trees. Construct a huffman code for the following data.

character	A	B	C	D	E	F
probability	0.30	0.30	0.13	0.12	0.10	0.05

Encode DAD using Huffman encoding.

- b) Define backtracking. Apply backtracking to solve the following instance of the subset sum problem $S = \{3, 5, 6, 7\}$ and $d=15$.
- c) Distinguish between branch and bound method and backtracking method.
8. a) Apply Dijkstra's single source shortest path algorithm on the following graph to find the shortest path by considering node s as the source.



- b) Define the minimum cost spanning tree(MST). Explain the Prim's algorithm to construct minimum cost spanning tree.
- c) Outline the Bellman-Ford algorithm to compute the shortest path.